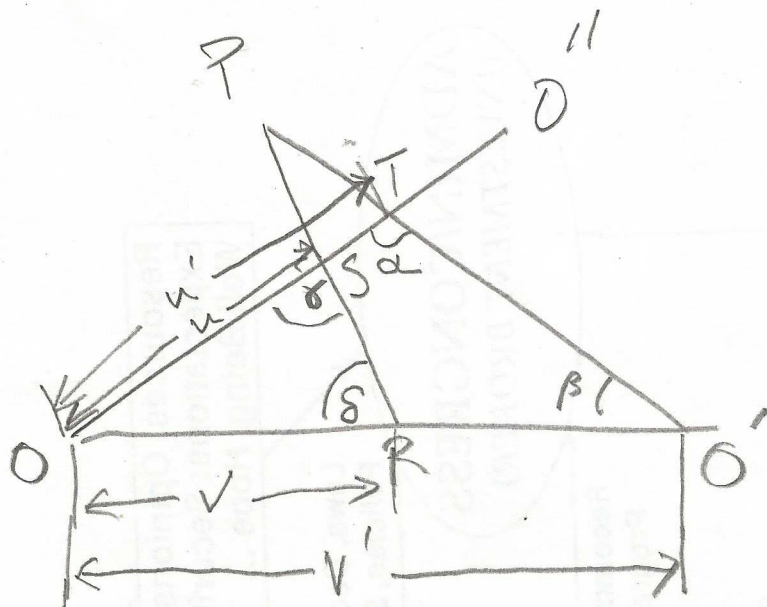
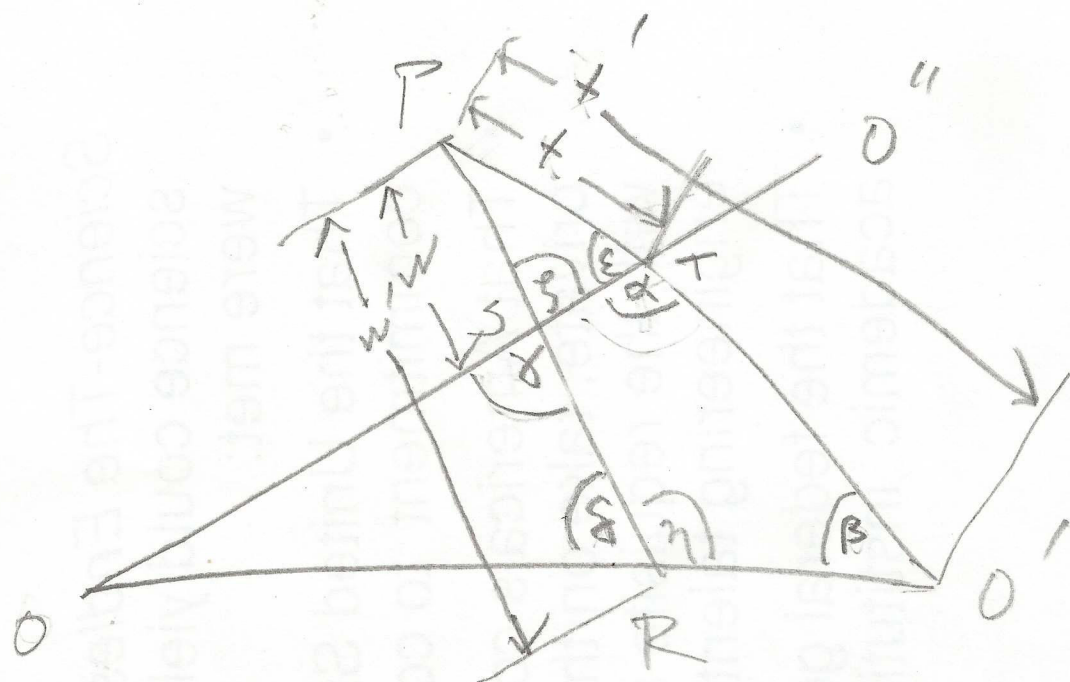




$$\frac{\sin \alpha}{\sin \beta} = \frac{v'}{u'}, \quad \frac{\sin \gamma}{\sin \delta} = \frac{v}{u}$$

$$\frac{u'}{\sin \beta} = \frac{v'}{\sin \delta} \Rightarrow$$

$$\frac{u}{\sin \delta} = \frac{v}{\sin \gamma} \Rightarrow$$



$$A = \frac{w/w'}{x/x'} = \frac{w/x}{w'/x'} = \frac{\sin \epsilon}{\sin \zeta} \cdot \frac{\sin \eta}{\sin \beta}$$

mult. top + bottom by $\frac{w'}{x}$

$$\frac{w}{\sin \epsilon} = \frac{x}{\sin \zeta} \Rightarrow \frac{w}{x} = \frac{\sin \epsilon}{\sin \zeta}$$

$$\frac{w'}{\sin \beta} = \frac{x'}{\sin \eta} \Rightarrow \frac{w'}{x'} = \frac{\sin \beta}{\sin \eta}$$

$$A = \frac{\sin \alpha}{\sin \delta} \cdot \frac{\sin \delta}{\sin \beta} = \frac{\sin \alpha}{\sin \beta} \cdot \frac{\sin \delta}{\sin \delta}$$

since $\alpha = \pi - \epsilon$, $\delta = \pi - \eta$, and $\sin(\pi - \theta) = \sin \theta$
and $\zeta = \delta$

As shown in the first figure,

$$A = \frac{v'}{u'} \cdot \frac{1}{\frac{v}{u}} = \frac{v'/u'}{v/u}$$

$$= \frac{\cancel{u}v}{u'/\cancel{v}'} = \frac{u/u'}{v/v'}$$

multiply top and bottom by $\frac{v}{u'}$